

MIP_PROMPT_LIBRARY_V2_PART1_DISCOVERY.md

Mainframe Intelligence Platform

Enterprise Prompt Library V2

Part 1 - Discovery Prompts

Version: 3.0

Status: Enhanced Enterprise Discovery Framework

Purpose

Discovery is the foundation of MIP.

A modernization initiative should never begin with code conversion.

It should begin with deep understanding.

The objective of discovery is not only to identify what exists, but to build a resilient, explainable, and future-ready understanding of the enterprise ecosystem.

Discovery must capture:

- Technical assets
- Business assets
- Operational assets
- Data assets
- Security assets
- Integration assets
- Infrastructure dependencies
- Execution patterns
- Failure points
- Recovery mechanisms
- Business criticality
- Modernization readiness

The goal is to create a complete enterprise knowledge foundation capable of supporting:

- Modernization
- Cloud Migration
- API Enablement
- Event-Driven Architecture
- AI-Assisted Engineering
- Impact Analysis

- Risk Assessment
 - Compliance Auditing
 - Operational Resilience
 - Digital Transformation
-

Discovery Design Principles

Evidence First

All findings must be traceable to source evidence.

System Thinking

Applications must be analyzed as interconnected systems rather than isolated programs.

Business Context Awareness

Discovery should identify technical assets and their relationship to business capabilities.

Resilience-Oriented Discovery

Capture information required to understand:

- Single Points of Failure
 - Recovery Paths
 - Operational Dependencies
 - Critical Business Processes
 - Disaster Recovery Risks
-

Graph-Ready Discovery

All discovered assets should be suitable for future knowledge graph construction.

AI-Augmented Discovery

Apply modern analysis techniques where appropriate:

- Static Analysis
- Dependency Analysis
- Pattern Recognition

- Semantic Clustering
 - Business Capability Mapping
 - Execution Flow Reconstruction
 - Data Lineage Discovery
 - Risk Scoring
-

Prompt 01

Enterprise Repository Discovery

File

prompts/discovery/repository_scan.md

Purpose

Perform a complete enterprise repository scan and create a comprehensive inventory of all technical, operational, and business artifacts.

Context

Large enterprise repositories often contain significantly more than source code.

Discovery should identify:

- COBOL Programs
- JCL
- PROCs
- Copybooks
- DB2 SQL
- VSAM Definitions
- CICS Definitions
- IMS Definitions
- MQ Definitions
- REXX Scripts
- Easytrieve Programs
- Scheduler Definitions
- Control Cards
- Utilities
- Configuration Files
- API Specifications
- Documentation
- Architecture Diagrams
- Operational Runbooks
- Test Assets
- Deployment Scripts

Inputs

Repository root folder.

Instructions

Analyze the repository recursively.

Identify:

- Source Artifacts
- Configuration Artifacts
- Operational Artifacts
- Infrastructure Artifacts
- Documentation Assets
- Test Assets
- Security Assets

Capture:

- File Types
- Naming Standards
- Folder Structures
- Ownership Indicators
- Environment References
- Version Indicators

Detect:

- Unknown File Types
- Duplicate Assets
- Orphaned Assets
- Dead Assets
- Archived Components
- Missing Documentation

Generate:

- Repository Heat Maps
- Artifact Distribution Analysis
- Technology Footprint Analysis
- Repository Health Assessment

Expected Output

Executive Summary

Repository Statistics

Artifact Type	Count
---------------	-------

Technology Landscape

Repository Inventory

| Type | Name | Path | Description |

Repository Health Assessment

Discovery Risks

Modernization Opportunities

Recommendations

Constraints

Do not infer unsupported business functionality.

Use observable evidence only.

Success Criteria

Every repository artifact is cataloged, classified, and traceable.

Example Usage

Analyze the repository under mfcodes and produce a complete enterprise artifact inventory.

Review Checklist

- All folders scanned?
- Unknown files identified?
- Duplicate assets identified?
- Repository health assessed?
- Technology footprint generated?

Prompt 02

COBOL Program Discovery and Execution Intelligence

File

prompts/discovery/program_inventory.md

Purpose

Build a complete inventory of COBOL programs and identify execution architecture, dependencies, complexity, and modernization readiness.

Context

Programs are the primary execution units of most mainframe systems.

Discovery should reveal:

- What programs exist
- How they interact
- Which programs are critical
- Which programs are obsolete
- Which programs are modernization candidates

Inputs

COBOL source files.

Instructions

Extract:

- PROGRAM-ID
- Author
- Compilation Options
- Called Programs
- Calling Programs
- Dynamic Calls
- Copybooks
- Files Referenced
- Tables Referenced
- MQ Usage
- CICS Usage
- IMS Usage
- External Interfaces
- APIs
- Batch Dependencies

Capture:

- Cyclomatic Complexity
- Structural Complexity
- Program Size
- Maintainability Indicators
- Technical Debt Indicators

Identify:

- Root Programs

- Utility Programs
- Shared Components
- Dead Programs
- Circular Dependencies
- High-Risk Programs

Generate:

- Dependency Maps
- Program Risk Scores
- Modernization Readiness Scores

Expected Output

Program Inventory

| Program | Path | Type | Complexity | Risk |

Dependency Summary

Root Program Candidates

Shared Components

Dead Code Candidates

Modernization Readiness Assessment

Missing Information

Constraints

Use source code evidence only.

Success Criteria

All COBOL programs are represented with dependency and risk intelligence.

Example Usage

Analyze all COBOL members and generate an enterprise execution inventory.

Review Checklist

- PROGRAM-ID extracted?
- Calls identified?
- Dynamic calls identified?
- Complexity measured?
- Root programs identified?

- Risk scores generated?
-

Prompt 03

Batch Processing and Operational Flow Discovery

File

prompts/discovery/jcl_inventory.md

Purpose

Build a complete inventory of batch jobs, operational workflows, execution chains, and recovery dependencies.

Context

JCL often represents the true operational architecture of the enterprise.

Understanding execution chains is critical for resilience and modernization.

Inputs

JCL members.

Instructions

Extract:

- JOB Names
- Steps
- EXEC Statements
- Procedures
- Utilities
- Datasets
- GDGs
- Sort Operations
- Scheduler References
- Restart Logic
- Recovery Logic

Identify:

- Batch Chains
- Job Dependencies
- Trigger Relationships
- Critical Processing Windows
- Long Running Jobs

- Failure Recovery Paths

Capture:

- Input Dependencies
- Output Dependencies
- Operational Sequences
- Scheduling Constraints

Generate:

- Batch Execution Graphs
- Critical Path Analysis
- Operational Risk Assessment
- Recovery Dependency Maps

Expected Output

Job Inventory

| Job | Step Count | Programs | Datasets |

Execution Summary

Batch Chain Analysis

Critical Path Analysis

Recovery Dependency Analysis

Operational Risk Assessment

Constraints

Do not infer unsupported business meaning.

Success Criteria

Every batch job and execution dependency is represented.

Example Usage

Analyze all JCL members and build an enterprise execution architecture.

Review Checklist

- Steps identified?
- Programs identified?
- Procedures identified?
- Scheduler references identified?

- Recovery logic identified?
 - Critical paths identified?
-

Prompt 04

Enterprise Data Structure Discovery

File

prompts/discovery/copybook_inventory.md

Purpose

Build a complete inventory of enterprise data structures and identify shared business information models.

Context

Copybooks often contain decades of embedded business knowledge.

Inputs

Copybook source files.

Instructions

Extract:

- Copybook Name
- Record Structures
- OCCURS Clauses
- REDEFINES Clauses
- Key Fields
- Data Types
- Field Lengths
- Validation Rules
- Business Codes
- Reference Data

Identify:

- Shared Structures
- Canonical Data Models
- Sensitive Data Fields
- PII Indicators
- Financial Data Indicators
- Regulatory Data Indicators

Generate:

- Data Structure Catalog
- Reuse Analysis
- Data Complexity Analysis
- Sensitive Data Assessment

Expected Output

Copybook Inventory

| Copybook | Records | Key Fields |

Structural Complexity Report

Shared Data Models

Sensitive Data Assessment

Reuse Analysis

Constraints

Do not invent business meaning.

Use evidence-based classification.

Success Criteria

All copybooks represented consistently and analyzed for enterprise significance.

Example Usage

Analyze copybooks and identify shared enterprise data structures.

Review Checklist

- Structures extracted?
 - OCCURS identified?
 - REDEFINES identified?
 - Sensitive fields identified?
 - Reuse measured?
-

Prompt 05

Enterprise Database Dependency Discovery

File

prompts/discovery/db2_dependency_discovery.md

Purpose

Identify all database dependencies and construct foundational data lineage intelligence.

Context

Database dependencies are critical for modernization, compliance, and impact analysis.

Inputs

COBOL source files containing EXEC SQL.

Instructions

Extract:

- Tables
- Views
- Aliases
- Synonyms
- Cursors
- Joins
- CRUD Operations
- Host Variables
- Stored Procedures
- Triggers
- Packages

Capture:

- Read Patterns
- Write Patterns
- Update Frequencies
- Shared Data Usage

Identify:

- Critical Tables
- Shared Tables
- Master Data
- Reference Data
- High-Risk Tables

- Compliance-Sensitive Tables

Generate:

- Dependency Matrix
- Data Flow Analysis
- Data Lineage Foundations
- Risk Assessment

Expected Output

Table Inventory

| Table | Operation | Program |

DB2 Dependency Matrix

Shared Tables

Critical Tables

Data Lineage Summary

High-Risk Tables

Constraints

Use SQL evidence only.

Success Criteria

All database dependencies are represented and traceable.

Example Usage

Analyze EXEC SQL statements and generate a DB2 dependency report.

Review Checklist

- Tables identified?
 - Operations classified?
 - Shared tables identified?
 - Critical tables identified?
 - Lineage foundations generated?
-

Prompt 05A

Root Program Discovery

File

prompts/discovery/root_program_discovery.md

Purpose

Identify true enterprise entry points.

Context

Many organizations do not know which programs actually initiate business processing.

Inputs

Program Inventory

JCL Inventory

CICS Definitions

Scheduler Definitions

Instructions

Identify:

- Batch Drivers
- Online Drivers
- Scheduler Entry Points
- Utility Entry Points
- Service Entry Points

Generate:

- Root Program Catalog
- Entry Point Rankings
- Dependency Reach Analysis

Expected Output

Root Program Inventory

Entry Point Analysis

Dependency Reach Report

Success Criteria

All execution roots identified.

Prompt 05B

Business Capability Discovery

File

prompts/discovery/business_capability_discovery.md

Purpose

Discover business capabilities embedded within the application landscape.

Context

Modernization should align with business capabilities rather than technical boundaries.

Inputs

Programs

Jobs

Copybooks

Documentation

Instructions

Apply:

- Semantic Analysis
- Naming Pattern Analysis
- Functional Clustering
- Dependency Clustering

Identify:

- Business Domains
- Business Capabilities
- Capability Ownership
- Capability Dependencies

Generate:

- Capability Map
- Domain Map
- Capability-to-System Matrix

Expected Output

Business Capability Catalog

Domain Architecture

Capability Dependency Analysis

Success Criteria

Business capabilities become visible and traceable.

Prompt 05C

Resilience and Operational Risk Discovery

File

prompts/discovery/resilience_discovery.md

Purpose

Identify operational resilience characteristics and systemic risks.

Context

Modern enterprises require resilient architectures.

Inputs

Programs

JCL

Datasets

Database Dependencies

Operational Assets

Instructions

Identify:

- Single Points of Failure
- Critical Dependencies
- Recovery Dependencies
- Restart Mechanisms
- Operational Bottlenecks
- Legacy Risks
- Unsupported Components

Assess:

- Availability Risks
- Recovery Risks
- Scalability Risks
- Security Risks

Generate:

- Resilience Score
- Risk Heat Map
- Recovery Readiness Assessment

Expected Output

Resilience Assessment

Risk Heat Map

Recovery Dependency Analysis

Recommendations

Success Criteria

Operational resilience posture is measurable and explainable.

Part 1 Completion Criteria

Upon completion of these prompts, MIP should be capable of answering:

- What exists?

- Where does it exist?
- How many artifacts exist?
- Which programs execute?
- Which jobs orchestrate execution?
- Which copybooks define data?
- Which DB2 tables are used?
- What are the root programs?
- What are the critical business capabilities?
- What are the operational dependencies?
- What are the resilience risks?
- What are the modernization candidates?
- What are the recovery paths?
- What are the single points of failure?
- Which assets are business critical?
- Which assets are cloud migration candidates?

Only after these questions are answered should parsing, metadata extraction, graph construction, and modernization intelligence begin.

End of Part 1.